



Quantification of NO in the post-flame region of laminar premixed ammonia/hydrogen/nitrogen-air flames using laser induced fluorescence

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ABSTRACT

Ammonia-based fuels have been identified as a promising alternative as zero-carbon energy carriers due to their high energy density and simpler logistics compared to hydrogen. As a disadvantage, the presence of fuel-bound nitrogen can lead to order of magnitude higher emissions of undesired nitric oxide (NO), nitrogen dioxide (NO₂) and nitrous oxide (N₂O) compared to more conventional fuels. Presently, chemical kinetics schemes for the combustion of ammonia and ammonia blends show large variations in the prediction of NO and there is a lack of quantitative experimental data to validate and optimize these reaction mechanisms. This paper presents measurements of NO in the product gases of laminar premixed NH₃/H₂/N₂ air flames on a flat-flame burner for 4 different ammonia decomposition ratios and over a range of equivalence ratios using laser-induced fluorescence in the NO A-X (0,1) system. A linear calibration approach based on the addition of NO to a lean premixed CH₄ flame is presented. Initial signal treatment includes the correction of laser absorption, fluorescence absorption (signal trapping) and fluctuations in laser energy. The LIF signals are corrected for changes in the Boltzmann fraction, line overlap, number density, and quenching between calibration and measurement, which requires knowledge of the local temperature and mole fractions of the main species. Temperature measurements using N₂ thermometry, where a theoretical N₂ Raman spectrum is fitted to an experimental N₂ Raman signal, excited by a 532 nm cw laser, allow characterization of the local near-adiabatic flame conditions as a function of operating conditions and adjustment of the signal corrections to the local temperature. Major species are extracted from 1-D simulations. The measured NO mole fractions are compared with five recent chemical kinetic schemes, which show good agreement for rich mixtures, however, a systematic underprediction of NO is found for stoichiometric and lean mixtures.

Novelty and significance

Emissions of NO are a major challenge for advancement of ammonia as a carbon-free fuel, yet very few measurements of NO levels in ammonia flames exist in the literature. In this paper, we present much needed quantitative experimental data on NO emissions from premixed NH₃/H₂/N₂-air flames using laser-induced fluorescence (LIF). Our diagnostic approach employs a linear calibration method based on the addition of NO to a lean CH₄ flame. Post-flame temperatures are measured by Raman spectroscopy to ensure accuracy of the local thermochemical states used in converting LIF signals to quantitative NO concentrations, accounting for variations in number density, electronic quenching, Boltzmann fraction, and the line overlap integral. Additionally, laser absorption and signal trapping are corrected using the spatial shape of the LIF signal. The presented data set can be used by modelers to refine chemical kinetic models with respect to NO emissions.

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1. Introduction

In recent years, the search for sustainable and environmentally friendly energy vectors has gained considerable momentum, driven by the urgent need to address global energy consumption and mitigate climate change. Ammonia (NH_3) has emerged as a compelling energy carrier, characterized by its high energy density compared to hydrogen and advantageous storage and transportation properties [1]. However, pure NH_3 is characterized by poor combustion properties that can be enhanced by partial decomposition of ammonia into hydrogen (H_2) and nitrogen (N_2), which was already demonstrated in 1967 by Verkamp et al. [2] and Neil and Bilger [3]. This partial decomposition can be optimized in relation to end use of the fuel blend and provides a lever to vary its combustion properties. Due to the presence of fuel-bound nitrogen and its oxidation pathways, a major challenge in ammonia combustion is the formation of nitric oxide (NO) and nitrogen dioxide (NO_2), which are highly toxic, and nitrous oxide (N_2O), which is a potent greenhouse gas. Thus, the prediction of emissions formation must be well understood to enable the development of future energy applications.

Recent experimental work focusing on flame propagation of (partially) dissociated fuel demonstrates a significant increase of laminar flame speed [4–6] and flammability limits [7] with increasing fuel decomposition. Furthermore, Wiseman et al. [8] showed drastically increased resistance to blow-out for turbulent bluff-body stabilized partially dissociated ammonia flames, which have similar laminar flame properties as methane. Measurements of extinction strain rates of laminar premixed $\text{NH}_3/\text{H}_2/\text{N}_2$ -air flames confirm the much higher resilience to strain-induced blow-out [9] with increasing H_2 percentage in the fuel, which underlines the suitability of these fuels in technical combustion applications.

Experimental data on NO formation and emissions in turbulent flames have been recently performed by Wang et al. [10] but such data in laminar flames is scarce, and state-of-the-art chemical kinetic schemes show significant differences in the prediction of NO [11]. Duynslaegher et al. [12] used molecular beam mass spectrometry to study the effects of initial hydrogen content, equivalence ratio, and pressure on ammonia combustion, showing that the effect of initial H_2 content on NO formation is small and that a reduction in the equivalence ratio compared to stoichiometric mixtures leads to significantly higher NO concentrations. In fact, observed post-flame NO mole fractions initially increase and then decrease slightly with increasing hydrogen mole fractions in the fuel for stoichiometric mixtures. However, an increase in NO emissions is expected as shown by Mei et al. [13] due to an increase of the O/H radical pool [14] when more H_2 is present in the fuel. Hayakawa et al. [15] recently measured the emission of a binary blend consisting of 70% NH_3 and 30% H_2 stabilized in a stagnation flow using a gas analyzer. The maximum NO emission was found to be around 8700 ppm at $\phi = 0.85$ which was in good agreement with the chemical kinetic scheme of Nakamura et al. [16]. In 2015, the same group measured NO emissions of pure ammonia flames on a nozzle-type burner using a gas analyzer [17] under elevated pressure, which showed that NO emissions decreased with increasing pressure. Brackmann et al. [18] investigated the flame structure of pure NH_3 -air flames obtaining profiles of OH, NH and NO using laser-induced fluorescence for three equivalence ratios, indicating that the remaining experimental uncertainties prevent accurate determination of rate constants in recent H/N/O models. Cosway et al. [19] investigated Bunsen flames with 65/35 NH_3/H_2 volume fraction at two equivalence ratios using planar laser induced fluorescence (PLIF) to image NO and OH reporting a higher production of NO in the NH_3/H_2 flames compared to the CH_4 reference flames. Thomas et al. [20] measured NO using saturated LIF in non-premixed counterflow flames for blends of NH_3 and H_2 ranging from 0% to 15% H_2 , concluding that none of the tested models accurately predicts the measured concentration profiles. Recently, El Baba et al. [21] employed calibrated

LIF to measure spatially resolved OH and NO profiles in low-pressure premixed $\text{NH}_3/\text{O}_2/\text{N}_2$ flames at three equivalence ratios. Their NO measurements showed good agreement in peak mole fractions with the kinetic model of Mei et al. [13] across all tested equivalence ratios. A fluorescence-free method for the measurement of NO was proposed by Zubairova et al. [22] to overcome typical challenges of the LIF technique. However, limitations of temporal and spatial resolution prevent the application of this method to turbulent flames.

None of the aforementioned studies focus on providing an experimental validation database for the NO emission of premixed partially dissociated ammonia flames. The present study fills this gap by quantitatively measuring local NO mole fractions in laminar flames fueled with four $\text{NH}_3/\text{H}_2/\text{N}_2$ blends over a range of equivalence ratios. A flat flame burner is used to provide well-defined conditions with respect to flow field, main species concentrations, temperature, and spatial uniformity [23]. NO mole fractions are measured quantitatively in the combustion products at 10 mm above the burner surface using quasi 1-D laser induced fluorescence. Factors that affect the NO-LIF signal are discussed and a calibration procedure - following a well-established approach based on the addition of NO [24–28] - is described, including corrections of the calibration factor due to changes in temperature and main species. The resulting NO mole fractions are compared to five recent chemical kinetic schemes.

2. Experimental setup

2.1. Burner and gas supply

A flat-flame burner (Holthuis and Associates) equipped with a stainless steel porous disk ($D = 60$ mm) and brass shroud ring was used in the experiments. The burner has well characterized boundary conditions, including temporal stability, radial uniformity, and axially slowly varying temperature and product gas composition [29]. The temperature of the burner body was monitored using a thermocouple, and water cooling at 20°C ensured that the burner never exceeded 30°C during the experiments. The flow velocity u of the fuel-air mixture was set to match its laminar flame speed, predicted by the Han mechanism [30]. An overview of all operating conditions is given in Table 1. The flow velocity of the shroud flow was kept at 50% of the main flow for all conditions. All gas flows were controlled by digital Bronkhorst mass flow controllers (MFC) with an accuracy of 0.5% of reading plus 0.1% of full scale. Note that in this study, the fuel blends are always identified by the mole fraction of H_2 in the fuel mixture. The other components can be derived from the H_2 content:

$$X_{\text{N}_2} = X_{\text{H}_2}/3$$

$$X_{\text{NH}_3} = 1 - (X_{\text{H}_2} + X_{\text{N}_2})$$

2.2. NO-LIF

An overview of the experimental setup for NO-LIF, absorption, and chemiluminescence measurements is shown in Fig. 1. The NO was excited in the A-X (0,1) system near 236 nm to avoid absorption of the laser by NH_3 [10,31,32]. The fluorescence was detected using a UV lens (100 mm F/2.8, Cerco) coupled to an intensifier IRO-X (LaVision) and imager M-Lite (LaVision). To suppress signal contributions from laser scattering, the detection system was equipped with a long-pass filter (ET240lp, AHF Analysetechnik). The filter blocks light below 237 nm ($\text{OD} > 5.0$) and its transmission rises sharply within 5 nm to 85%–93%. No signal bias is expected for all red-shifted transitions ($v'' > 2$) that are recorded in the experiments. The gating time of 500 ns efficiently suppresses any background signal contributions, for instance from chemiluminescence of excited NO_2^* or NH_2^* [33]. An intensifier gain of 85% provides a signal-to-noise ratio (μ/σ) of ≈ 19 at 10000 ppm NO.

The excitation wavelength of 236 nm was generated in a cascade process for third harmonic generation: A dye laser (Cobra Stretch,

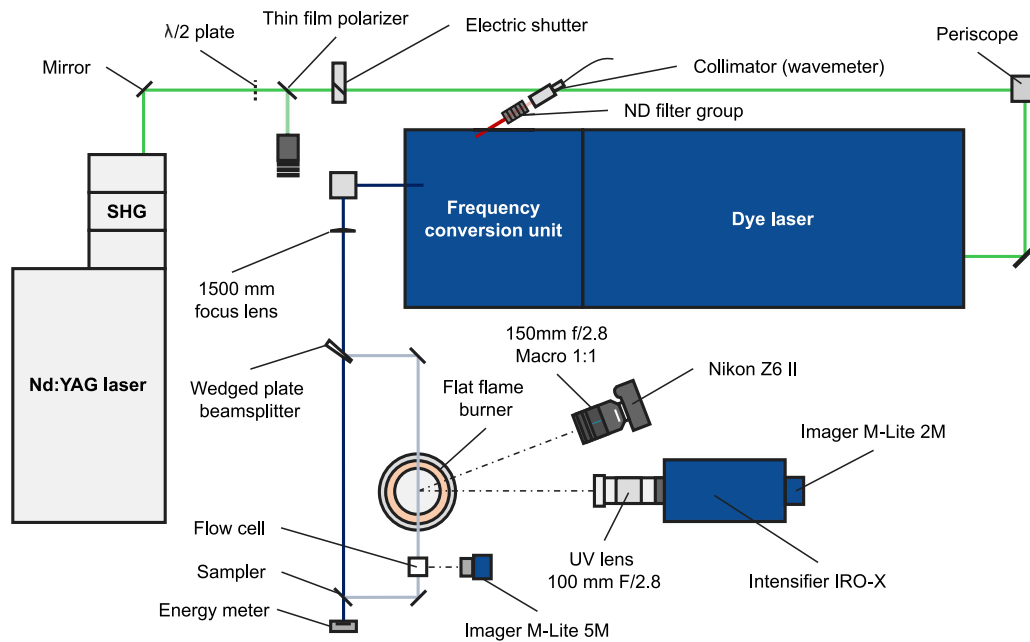


Fig. 1. Experimental setup (NO-LIF, absorption and chemiluminescence).

Table 1
Overview of operating conditions.

% H ₂	ϕ (-)	u (cm s ⁻¹)	% H ₂	ϕ (-)	u (cm s ⁻¹)
27	0.70	7.96	33	0.70	10.54
	0.75	9.63		0.75	12.81
	0.80	11.26		0.80	14.80
	0.85	12.85		0.85	16.69
	0.90	14.24		0.90	18.43
	0.95	15.43		0.95	19.88
	1.00	16.64		1.00	21.34
	1.05	17.82		1.05	22.70
	1.10	18.10		1.10	23.08
	1.15	17.73		1.15	22.58
	1.20	16.29		1.20	21.45
	1.25	15.96		1.25	20.31
	1.30	14.84		1.30	19.00
39	0.60	9.23	45	0.60	13.81
	0.65	12.18		0.65	17.78
	0.70	15.07		0.70	21.58
	0.75	17.75		0.75	25.34
	0.80	20.30		0.80	28.39
	0.85	22.65		0.85	31.78
	0.90	24.99		0.90	34.97
	0.95	27.07		0.95	37.90
	1.00	28.96		1.00	40.50
	1.05	30.43		1.05	42.48
	1.10	30.99		1.10	43.30
	1.15	30.39		1.15	43.06
	1.20	28.04		1.20	41.57
	1.25	27.29		1.25	39.45
	1.30	25.46		1.30	36.88
	1.35	23.69		1.35	34.29
	1.40	22.00		1.40	31.73

Sirah), operated with Pyridine 1, was pumped with the second harmonic of a Nd:YAG laser (Spotlight 1000.2, Innolas) at 532 nm. The fundamental of the dye (708 nm) was frequency doubled in a first BBO-crystal to obtain 354 nm. After doubling, sum frequency mixing in a second crystal of the remaining 708 nm beam and its second harmonic was used to obtain 236 nm. The pulse energy was adjusted by rotating a $\lambda/2$ plate (WPH10M-670, Thorlabs) in front of the crystals, thus influencing the efficiency of the subsequent frequency conversion process. The overlapping beams were separated by four Pellin-Broca prisms. The second harmonic was directed to a beam dump and the remains of the 708 nm beam were used to track and record the center

line excitation wavelength during the experiment by a wave meter (WS6-200, HighFinesse). Using a 1500 mm spherical lens, the laser beam intersected the burner centrally. After passing the focus lens, a wedged plate beam splitter (Thorlabs) was inserted to the beam path to reflect $\approx 1.6\%$ of the UV laser. This small portion was used for the LIF measurements and was sufficient to reach the onset of partial saturation in combination with the 1500 mm focus lens. In previous experiments we tested a 750 mm lens to estimate its suitability for measurements in turbulent flames, where spatial resolution is important. However, due to the ≈ 4 times higher local intensity, LIF saturation occurred much earlier and the SNR was $\approx 40\%$ lower compared to the 1500 mm lens. The remaining energy was guided to an energy meter (J-25MB-LE, Coherent) to monitor the pulse energies on a single-shot base. A flow cell filled with a solution of quinine and water (0.5 g/L) was used to measure the absorption of the laser radiation by NO and for an additional energy reference, which is less noisy compared to the energy meter that was operated at the lower end of its specifications. Parts of the laser beam before and after passing the flame were directed into the flow cell from opposite sides and at different heights. The fluorescence of the quinine in the visible spectral range was detected with a CMOS camera (Imager M-Lite 5M, LaVision) perpendicular to the beams. Verification of the signal linearity of the quinine-water solution was checked by an energy scan. Chemiluminescence images of all flames were recorded using a mirrorless camera (Z6 II, Nikon) combined with a macro lens (150 mm F/2.8 Macro 1:1, Irix). The camera was set to 1 s exposure time, an aperture of F/32, and ISO 640.

2.3. N₂-thermometry

An accurate knowledge of the thermochemical states for all calibration and measurement conditions is required to correct the fluorescence signal. In the present study, the gas composition was extracted from 1-D simulations using a freely propagating flame. However, even if the temperature could be taken from the same simulations, post-flame temperatures are expected to be below the adiabatic flame temperature due to heat transfer to the water-cooled burner matrix. Therefore, local temperatures were measured for all operating conditions using N₂-Raman thermometry. Following Lill et al. [34], N₂ Raman spectra were measured at the same height above the burner using a 532 nm cw laser and a theoretical spectrum is fitted to the experimental one to

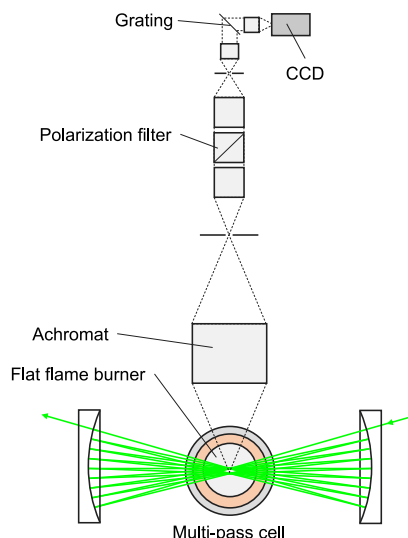


Fig. 2. Experimental setup for N_2 -thermometry by Raman scattering.

Table 2
Chemical kinetics schemes compared in this study.

Name	# of species	# of reactions	Ref.
Jiang	19	64	[38]
Han	20	130	[30]
Nakamura	33	232	[16]
Stagni	31	203	[39]
Wang	91	444	[40]

obtain the local temperature. A multi-pass cell was used, as shown in Fig. 2, which enhances Raman signals by a factor of ≈ 8 . The Raman scattering collected with an achromatic lens was passed through a polarization filter to transmit only s-polarized scattering, dispersed with a transmission grating (2380 lines/mm, Wasatch Photonics) and recorded with a CCD camera (ProEM+, Princeton Instruments). More details about the spectrometer are given by Dieter et al. [35] and Lill et al. [34]. To minimize readout noise, the exposure time of the CCD camera was adjusted to maximize signals without saturating the sensor. Background signals dominated by NO_2^* and NH_2^* in fuel-lean and fuel-rich flames [33], respectively, were recorded without the laser. By subtracting these flame-specific spectra for all cases considered, the background was removed efficiently.

2.4. 1-D simulations

1-D flame simulations were performed in Cantera [36] using a freely propagating flame configuration. A multi-component transport model was used and Soret diffusion was included in the calculations as this has a significant effect on the prediction of emissions [37]. For the comparison of NO predictions with experimental data, NO mole fractions were taken 10 mm downstream of the maximum temperature gradient to ensure that the comparison is conducted at homologous positions in the post flame region. Table 2 summarizes all the chemical kinetics schemes used in this study for comparison with experimental data. Table 2 also summarizes the number of species and reactions that are included in the mechanisms.

3. Fluorescence evaluation

3.1. Multi-level non-transient linear LIF model

The excitation line at 235.976 nm was used for single-photon excitation of NO because it is the strongest feature in the A-X (0,1) system. This absorption feature consists of several transitions contributing to

the fluorescence signal. The strongest signal contribution ($\approx 50\%$) is from the Q_1 (15.5) transition. Additional rather strong transitions ($> 5\%$ signal contribution) that are inevitably excited during the experiment are the R_2 (15.5), Q_{21} (9.5) and R_1 (9.5) transitions. The spectral width of the broadened absorption feature ϕ_A is in the same order as the laser line width ϕ_L in this specific experimental setup. This requires the overlap of the laser excitation and each contributing transition to be calculated separately. A common approach is to assume a constant spectral energy density across the absorption feature, which is not valid when the energy density changes rapidly over a small spectral region. Therefore, the line shape of the laser excitation must be well known and was characterized by performing an excitation scan. Fitting a theoretical NO excitation spectrum (Voigt profile) to the experimental one using LIFSIM [41] provided the laser line shape with a full width at half maximum (FWHM) of 0.3261 cm^{-1} .

Assuming weak laser excitation, the contribution to the fluorescence signal S_F from multiple laser-coupled transitions i is expressed by Eq. (1) in terms of NO mole fraction X_{NO} , number density n , length of probe volume L , pulse energy E_L , spectral FWHM of the laser $\Delta\nu_L$, detection efficiency η , Boltzmann fraction $f_{B,i}$, dimensionless spectral overlap fraction Γ_i , Einstein B-coefficient for induced absorption $B_{12,i}$ and fluorescence yield F_i .

$$S_F = X_{NO} n L \frac{E_L}{\Delta\nu_L} \eta \sum_i f_{B,i} B_{12,i} \Gamma_i F_i \quad (1)$$

Note that the dimensionless definition for the spectral overlap fraction Γ_i from Partridge and Laurendeau [42] is used here, where $\phi_{A,i}(\nu)$ is the shape of the absorption line normalized to unity on spectral integration and $L_L(\nu)$ the laser line shape normalized to its FWHM on spectral integration.

$$\Gamma_i = \int \phi_{A,i}(\nu) L_L(\nu) d\nu \quad (2)$$

The fluorescence yield F_i is given by $A_{23,tot}/(A_{23,tot} + Q_{tot})$. Here, $A_{23,tot}$ is the total Einstein A-coefficient of all detectable rovibronic fluorescence transitions and Q_{tot} the electronic quenching rate. The total Einstein A-coefficient is calculated from all A-X (0, ν'') (with $\nu'' \geq 2$) transitions j , where A-state energy transfer processes are neglected [41].

$$A_{23,tot} = \sum_j A_{23,j} \quad (3)$$

3.2. Calibration approach

For calibration in a thermochemical system with known NO concentrations, defined amounts of NO were doped into a lean CH_4 flame at $\phi = 0.8$. 1-D simulations of freely propagating flames (CRECK C₁-C₃ detailed reaction mechanism [43–45] including NO_x sub-mechanism [46]) show a nearly linear relationship between NO concentrations in the fresh gas and the post-flame region at low NO seeding levels (initial NO mole fractions from the CH_4 flame without NO addition were subtracted in this analysis). At higher NO seeding levels, significant deviations ($\approx 10\%$ at 5000 ppm) are predicted by the simulations. Note that the measured linearity extends to 7000 ppm, as presented later in Section 4.3.2. Assuming a linear relation, which holds for NO-seeding levels of up to 7000 ppm, the calibration factor C_C can be extracted from the measurement to directly link the signal intensity to NO concentrations as shown in Eq. (4).

$$S_F = C_C X_{NO,C} \quad (4)$$

According to the simulation results, the influence of NO addition on the laminar flame speed is below 2.5% when the NO addition is below 10000 ppm and the influence on the adiabatic flame temperature is negligible. The oxidizer consisted of a mixture of 21% O_2 and 79% N_2 and was supplied from gas cylinders. In a stepwise procedure, the N_2 flow was partially replaced by a mixture of 10% NO / 90% N_2 to obtain

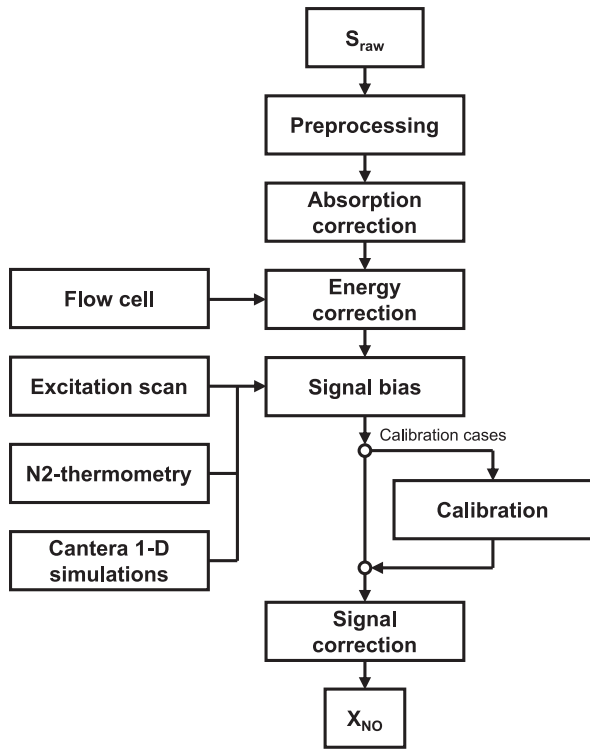


Fig. 3. Flow chart of the processing procedure including all signal sources.

0–10000 ppm NO in 1000 ppm steps. Signal contributions from native NO in the CH₄ flame are recorded without the addition of extra NO and subtracted from all signals.

The calibration factor is influenced by the factors shown in Eq. (1) and therefore depends on temperature T and main species concentrations $\bar{c}$. Accordingly, C_C is only valid when the gas composition and temperature are the same as during calibration experiments, which is typically not given in LIF experiments. Therefore, changes in the proportionality constant due to changes in T and $\bar{c}$, which have a strong influence on the number density, Boltzmann fraction, electronic quenching and line overlap, must be taken into account in the signal processing (see Sections 3.3.3 and 3.3.4).

3.3. Signal processing

Fig. 3 presents a flow chart that illustrates all signal processing steps that were applied to compute NO mole fractions from raw LIF signals. A total of ≈ 400 images were used for LIF signal processing at each condition and ≈ 100 images were used for background correction. During preprocessing, all images were background corrected and integrated in stripes of 80 px perpendicular to the laser and 16 px (≈ 0.63 mm) along the laser beam, providing a 1D signal profile for further data analysis. Absorption (Section 3.3.1) and energy correction (Section 3.3.2) are made using the 1D profiles. Afterwards, signals were averaged in time and space to obtain a single signal value for each measurement. A signal bias, containing information such as number density, electronic quenching rates, Boltzmann fractions and line overlap integral, was assigned to each individual condition (Section 3.3.3) and the calibration factor was then corrected according to these conditions (Section 3.3.4). To obtain corrected values, raw signals were divided by the corresponding correction values γ_i .

3.3.1. Absorption correction

The fluorescence signal can be significantly biased due to laser absorption by NO molecules, which reduces the local excitation intensity along the laser beam path through the product gases. In addition,

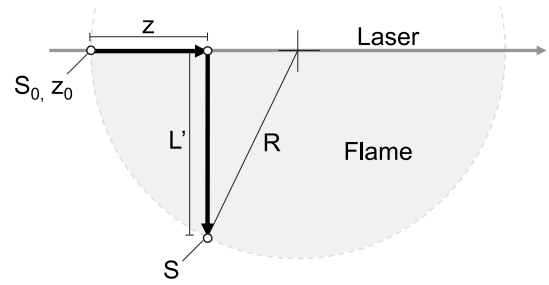


Fig. 4. Path lengths for laser absorption along laser axial coordinate z and signal trapping L' .

fluorescence photons can be reabsorbed on their way from the probe volume to the detector (signal trapping) as illustrated in Fig. 4. To correct for both effects, we used the spatial profile of the signal itself and assumed spatially homogeneous thermochemical conditions along most of the path (e.g. no change in radial direction of temperature and gas composition including NO) at 10 mm above the burner surface. Alternatively, the total laser absorption from the flow cell fluorescence could have been used. Since the effects of signal trapping are not captured by absorption measurements using the flow cell, we preferred the approach presented below, which allows correction for both effects.

The dependence of the local signal strength $S(z)$ in relation to the initial signal S_0 , where z is the axial coordinate along the laser beam, can be described by the product of two exponentially decaying waveforms as shown in Eq. (5). k_A and k_T are the absorption coefficients for laser absorption and fluorescence trapping, respectively. L' is the absorption length for signal trapping that is assumed to be limited to the product gases, where fluorescence photons can be reabsorbed by NO populated within the X-state at $v'' \geq 2$. In the present cases, L' can be approximated by Eq. (6), assuming a perfectly round shape of the post flame region. k_A and k_T are quantified by fitting Eq. (6) to the recorded LIF signal. Note that R is the radius of the product gases, which is at 10 mm above the burner different to the radius of the burner exit because the product gases are conically shaped (see 4.1). Thus, R is directly estimated using the NO-LIF signals.

$$\gamma_{TA}(z) = S(z)/S_0 = \exp(-k_A z) \exp(-k_T L') \quad (5)$$

$$L' = \sqrt{R^2 - (2R - z)^2} \quad (6)$$

3.3.2. Energy correction

For energy reference, the flow cell signals were shot-to-shot integrated in separate pixel windows to obtain two individual intensity values before and after the flame passage. Signal-energy relations were obtained through linear fitting of the LIF signals and flow-cell signals (before the flame, S_{FC}) of the image series. Since the slope p of the signal-energy relation slightly depends on the location z , this correction was applied to each position along the laser beam path. Signal values were then corrected to a reference flow cell signal $S_{FC,ref}$ as shown in Eq. (7). The correction value for each location $\gamma_E(z)$ is found by inserting the signal value S and the flow-cell signal S_{FC} .

$$\gamma_E(z) = \left(1 - \frac{p(z)}{S} [S_{FC} - S_{FC,ref}]\right)^{-1} \quad (7)$$

3.3.3. Signal bias

As shown in 3.1, the LIF signal is biased by a number of factors that can be corrected for if the thermochemical states are known. Therefore, main species mole fractions were taken from 1-D flame simulations at 10 mm downstream of the maximum temperature gradient using the Han mechanism. The agreement between the kinetic models (Table 2) for the prediction of the main species is very good, so the choice of

Table 3
Fit parameters for temperature-dependent NH_3 quenching cross section.

c_1	c_2	c_3	c_4
1168	79.8	-1124	-69.1

mechanism plays a minor role. Since adiabaticity cannot be assumed due to significant heat loss to the burner, temperatures based on the N_2 -thermometry measurements were used for this evaluation. The spectral line shape of the laser was characterized using an excitation scan and fitting a theoretical spectrum to the experimental data. Afterwards, all relevant factors were determined, as described in the following paragraphs.

Ground state Boltzmann fractions $f_{B,i}$ are a function of temperature and ground state energy only and were calculated using the temperature measured in the N_2 -experiments. The partition function was calculated using spectroscopic data from the LIFSIM database from Paul [47]. The number density n was computed directly from the gas composition and temperature using the ideal gas law.

The dimensionless overlap fraction Γ_i was calculated for each single transition, using the laser line shape obtained from the excitation scan and the absorption line shape determined by a Voigt profile that is obtained by convolution of a Lorentzian (collisional broadening) and Gaussian (Doppler broadening) distribution. Collisional broadening coefficients for the main species that are expected in the product gases are available for N_2 [48], O_2 and H_2O [49] and CO_2 [50,51]. Temperature-dependent electronic quenching cross sections for N_2 , O_2 , H_2O and CO_2 as well as NO self-quenching rates are available in the literature [52]. Quenching on NH_3 was also considered in the evaluation, although it is only expected at very rich conditions where ammonia slip might occur. Experimental quenching cross sections from Furlanetto et al. [53] were fitted using a function proposed by Settersten et al. [52] as shown in Eq. (8). The results for the fit parameters are summarized in Table 3.

$$\sigma(T) = c_1 \exp(c_2/T) + c_3 \exp(-c_4/T) \quad (8)$$

Temperature dependent quenching rates q_i were evaluated according to Eq. (9), where σ is the temperature dependent quenching cross section, μ is the reduced mass, and k_B is the Boltzmann constant. The total quenching rate Q_{tot} for species k is given in Eq. (10).

$$q = \sigma(T) / \sqrt{\frac{\pi \mu k_B T}{8}} \quad (9)$$

$$Q_{\text{tot}} = \sum_k q_k p X_k \quad (10)$$

Main quenchers in the present experiments are CO_2 and H_2O for the CH_4 flame and H_2O for the $\text{NH}_3/\text{H}_2/\text{N}_2$ flames, respectively.

Further spectroscopic data such as transition energies and Einstein-coefficients were taken from the LIFSIM database [41] and are based on the term values of the NO A and X electronic states from Paul et al. [47].

3.3.4. Calibration and signal correction

The calibration factor C_C is extracted from the calibration measurements and contains all parameters that are required to describe the LIF signal (see Eq. (1)):

$$C_C = nL \frac{E_L}{\Delta\nu_L} \eta \sum_i f_{B,i} B_{12,i} \Gamma_i F_i \quad (11)$$

Since some of the parameters in Eq. (11) are influenced by thermochemical states, a correction must be applied to C_C when thermochemical states change, which is the case for all measurements except the calibration measurement. The final equation to compute the NO mole fractions is shown in Eq. (12), where S_M is the signal obtained during the measurements and γ_C is the correction term for C_C (the indices C and M denote calibration and measurement conditions, respectively).

$$X_{\text{NO}} = \frac{S_M}{\gamma_C C_C} \quad (12)$$

Assuming that the length of the probe volume L , the FWHM of the laser excitation $\Delta\nu_L$, and the detection efficiency remain constant between measurement and calibration, the correction value γ_C is given by Eq. (13). Note that the pulse energy E_L is not considered in the equation since all signals were energy normalized beforehand and therefore $E_{L,M} = E_{L,C}$.

$$\gamma_C = C_M / C_C = \frac{n_M \sum_i (f_{B,i} B_{12,i} \Gamma_i F_i)_M}{n_C \sum_i (f_{B,i} B_{12,i} \Gamma_i F_i)_C} \quad (13)$$

3.4. Uncertainty estimation

Experimental uncertainties were analyzed by manipulating the processing parameters and evaluating the deviation of the resulting mole fraction of NO. In this analysis, the case $33\% \text{ H}_2 / \phi = 0.8$ was used exclusively. Although the temperature was measured with N_2 -thermometry and compared to laminar flame simulations, we assumed an uncertainty of 2.5%, leading to an error of 1% in the experimentally determined mole fractions. The rather low sensitivity to temperature is probably due to counteracting trends, for example f_B , which increases, and n , which decreases with increasing temperature.

Especially with narrow band laser excitation, small changes in the excitation wavelength due to experimental drifts have a rather large influence on the results. The wavelength fluctuations are in the sub-picometre range and the sensitivity to the signal was checked using the excitation scan. Since signal changes are very small for the observed wavelength fluctuations, no significant error is expected in the results. Furthermore, the influence of uncertainties in the laser spectral line shape obtained from the excitation scan was tested and is negligible, even if the FWHM of the laser were 10% larger than estimated (which is unlikely considering the very good agreement between experiment and simulation as shown in 4.2.2). However, larger experimental uncertainties are linked to the quenching cross sections, that were studied by a variety of authors. As shown by Settersten et al. [52], the quenching cross sections of the main quenchers in the present work, namely H_2O and CO_2 , measured and modeled by a number of experimentalists, vary significantly. The quenching cross sections measured and modeled by Settersten et al. lie above other models in the temperature range of interest for the present set of experiments (≈ 1700 – 2100 K), so we have assumed an uncertainty of $\sigma_{\text{H}_2\text{O}}$ of -10% (higher quenching cross sections are not expected based on available literature data). For σ_{CO_2} we have included $\pm 5\%$ into our uncertainty analysis, which is rather conservative given the fairly good agreement of the models by Settersten et al. and Paul [54] in the considered temperature range. These deviations lead to experimental uncertainties of $+2.3\%$ / -6% , including an additional 0.5% in both directions from all other quenchers, which are rather insensitive even for larger perturbations. Uncertainties of the broadening parameters were assumed to be $+5\%$ / -9% based on the specification of DiRosa et al. [49] for O_2 and H_2O . However, the impact on the results was minor and therefore this effect has been neglected in the total uncertainty estimation. Total systematic experimental uncertainties are estimated to be $+3.3\%$ / -7% .

Some flames with very low reactivity, in particular rich mixtures with $\phi \geq 1.2$ tended to be significantly lifted from the burner surface, which brought the reaction zone closer to the probe volume. Since the NO formation in rich mixtures is also affected by the comparably slow Zel'dovich mechanism [55,56], NO profiles are not entirely flat downstream the reaction zone and the direct comparison of experimental and numerical data might be affected by deviation of the intended and actual probing position. Therefore, the influence of flame lift-off has been included in the experimental error bars by taking the relative change of the NO mole fraction from 10 to 7 mm after the largest temperature gradient as potential uncertainty, which would represent a flame that is lifted by 3 mm.

Normalized temperature ($\Theta = (T - T_0)/(T_{\text{ad}} - T_0)$) profiles were used to test if all gases were in thermal equilibrium at the probed position.

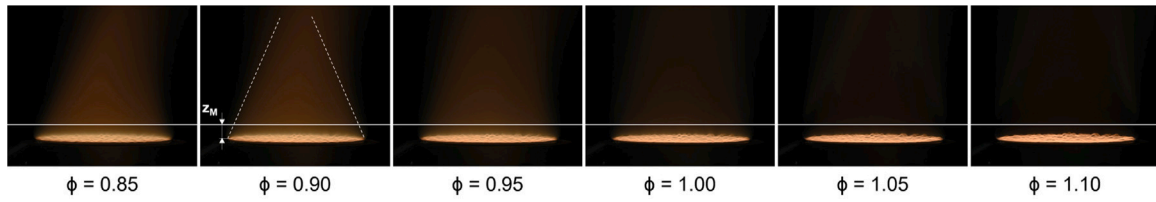


Fig. 5. Chemiluminescence images for selected 39% H₂ cases. The horizontal line indicates the axial position of the LIF measurement, z_M . For $\phi = 0.90$ the lines indicate the conical shape of the exhaust gas stream.

The spatial temperature gradients around the probed position are close to zero for all conditions and therefore thermal equilibrium is assumed. Reducing the probe position from 10 to 7 mm does not affect these results.

4. Results and discussion

4.1. Chemiluminescence

Fig. 5 shows chemiluminescence images of selected cases with 39% H₂ content to illustrate the change in flame appearance during the transition from lean to rich mixtures. The visible luminescence in the reaction zone and post-flame region can be attributed to the chemiluminescence of NO₂* and NH₂* in the lean and rich mixtures, respectively [33]. The luminosity in the post-flame region decreases considerably when transitioning from lean to rich flames due to the absence of NO₂ in the post-flame region at rich conditions. NO₂ mole fractions 10 mm downstream of the preheat zone taken from 1-D simulations (Han mechanism) show that some NO₂ is present at $\phi = 1$, which disappears completely at $\phi \geq 1.05$ and is in good agreement with the recorded chemiluminescence in Fig. 5.

4.2. Preparatory measurements

4.2.1. N₂-thermometry

Fig. 6 shows the temperature obtained from the N₂-fit normalized to the adiabatic flame temperature (Han mechanism) that was extracted from 1-D freely propagating flame simulations. Temperature predictions from the different mechanisms are within $\pm 0.6\%$.

The temperatures obtained from the fitting of the temperature-dependent N₂ Raman spectra are 0%–4% below the adiabatic flame temperature. This indicates some heat loss from the flame to the environment. Lean and far-rich mixtures are closer to the adiabatic flame temperature, and the deviations for stoichiometric and slightly rich flames are somewhat larger. These trends agree well with the observations of Weigand et al. [23], however our measurements are generally closer to T_{ad} because the flow velocity relative to the laminar flame speed is much higher in our experiments ($u = s_L$), increasing the flame distance to the burner matrix and therefore reducing heat loss. The differences of T/T_{ad} may also be due to small deviations between the laminar flame speed calculations and the true flame speed. For the correction of the LIF signals, the temperature in each case is obtained from a smoothing spline fit of T/T_{ad} to ϕ as shown in Fig. 6.

4.2.2. Excitation and energy scan

An excitation scan is used to estimate the spectral line shape of the laser excitation by fitting a theoretical excitation spectrum to the experimental one. As outlined in Section 3.3, this is required to accurately calculate the spectral overlap between laser and absorption line, which depends on temperature (Doppler broadening) and gas composition (collisional broadening). A mixture containing 33% H₂ at $\phi = 0.8$ was used in the excitation scan with a concentration of ≈ 10000 ppm NO. The wavelength was tuned continuously and tracked with the wave meter. A total of ≈ 1350 images were used in the data analysis, and the data at the burner center line position was used for evaluating

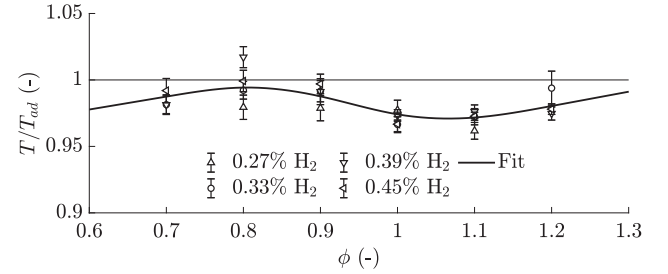


Fig. 6. Relative deviation from adiabatic flame temperature for all cases between $\phi = 0.7 - 1.2$.

the laser line shape. The scan was performed between 235.952 nm and 236.033 nm ($42366.9 - 42381.4$ cm⁻¹) in the spectral vicinity of the strongest NO peak at ≈ 235.976 nm in this branch. A theoretical spectrum was obtained from LIFSIM, using the temperature from the N₂-thermometry Raman measurements (1880 K) and main species from 1-D flame simulations as input. The results are shown in Fig. 7(a). The FWHM of the instrument line shape (Voigt profile) is 0.3261 cm⁻¹.

An energy scan is used to check the linearity between signal and excitation energy. The laser wavelength was fixed at 235.976 nm to probe the transitions that are described in Section 3.1. Fig. 7(b) depicts signal and pulse energy of an energy scan performed at the same flame conditions as in the excitation scan. The energy was tuned by slowly rotating the $\lambda/2$ -plate that is located before the frequency conversion crystals. An empirical model was fitted to the data to assess the linearity of the signal Eq. (14). The independent parameters are α and E_{sat} , a scaling factor and the saturation energy, respectively. The model shows very good agreement, although being very simple and not featuring any actual spectroscopic nor experimental values. At pulse energies of around 800 μ J, the onset of partial saturation (recognizable deviation from saturation fit to linear fit) is visible. We therefore limited the pulse energy for the experiments to ≈ 700 μ J, which corresponds to an energy in the probe volume $\bar{E}_{PV}$ of ≈ 12 μ J. The gray area in Fig. 7(b) represents the mean energy and its fluctuations (1σ) during the NO-LIF quantification, indicating that saturation is avoided during the experiments.

$$S(E) = S_0 + \alpha \frac{1}{1 + E_{sat}/E} \quad (14)$$

4.3. No mole fractions

4.3.1. Absorption

Fig. 8 shows the performance of the absorption correction that is described in Section 3.3.1. The steep spatial gradients dS/dz at both ends of the raw signal S_{raw} were used to identify the radial extension of the homogeneous post-flame region. Inside the homogeneous post-flame region, the correction function γ_{TA} was fitted to the signal shape and the corrected signal was generated as shown in Eq. (5). Data for the evaluation of post-flame NO mole fractions was extracted in a narrower region with a constant offset (10 mm from the edges of the homogeneous zone). Note that raw and corrected signals in Fig.

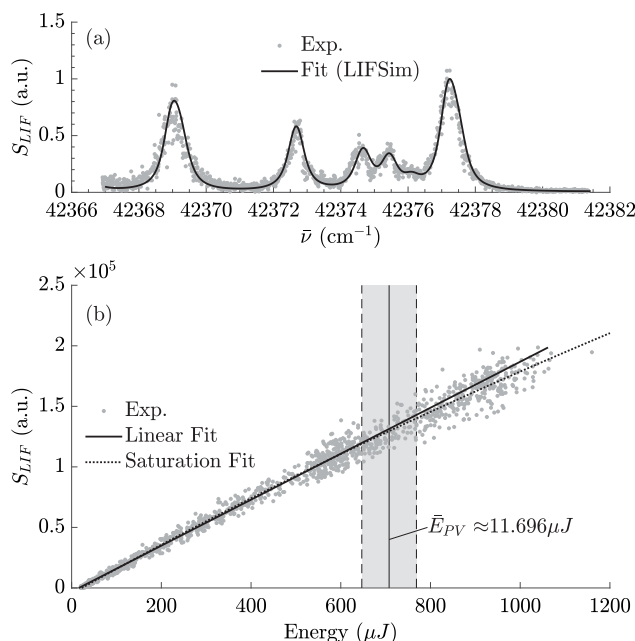


Fig. 7. Excitation scan (a) and energy scan (b). Both scans are recorded downstream the reaction zone of a 33% H_2 flame at $\phi = 0.8$. The gray area marked in (b) represents the mean energy and its fluctuations (1σ) that was used in all LIF experiments. Note that the energy that was measured by the energy meter is about two orders of magnitude higher than the energy in the probe volume, which was estimated to be around $12\mu\text{J}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

8 are normalized to $S_0 = S_{\text{raw}}(z = 0)$ and that γ_{TA} is fitted to time-averaged signals. After absorption correction, all single-shot LIF signal values inside the evaluation window were averaged and used for further analysis.

4.3.2. Calibration

Fig. 9 shows the results of the calibration measurements in the lean CH_4 flame. The linear regression was limited to a NO concentration of 7000 ppm in the data evaluation, as an onset of non-linearity can be observed at higher concentrations. This non-linearity is expected when large amounts of NO are added to the flame (see 3.2). Including NO mole fractions > 7000 ppm in the evaluation would alter the calibration factor C and distort the measured NO mole fractions. Following this procedure, the calibration factor of $C_C \approx 19.2$ counts/ppm was determined with a R^2 of > 0.99 . In order to reduce the scale errors at low flow rates over the wide range of mole fractions to be calibrated, we used two different mass flow controllers with different maximum volume flow rates for the addition of the NO/N_2 mixture. Switching between the mass flow controllers shows the two data points at 4000

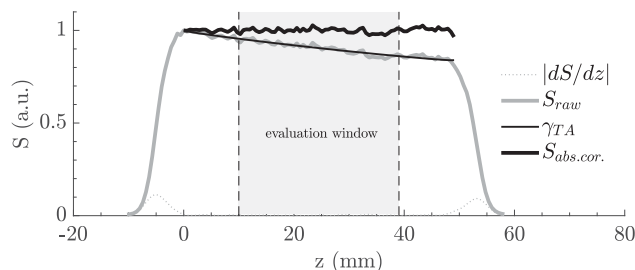


Fig. 8. Correction of laser absorption and signal trapping. The signals used for the evaluation of NO mole fractions are limited to the evaluation region located in the core of the product gases.

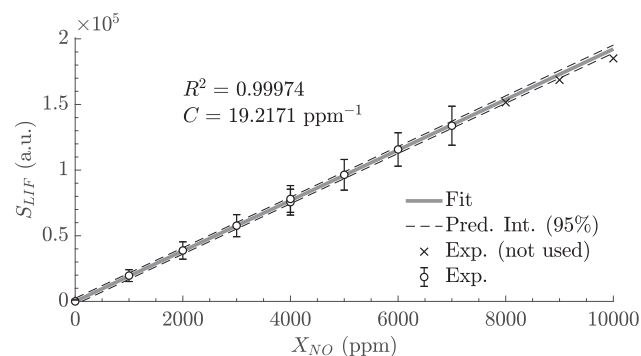


Fig. 9. NO-LIF calibration. The error bars represent 2σ signal fluctuations. Data points out of the linear range that were not used to evaluate C are shown as well.

ppm. The prediction intervals shown in Fig. 9 are included in the total uncertainty of the subsequently measured NO mole fractions.

4.3.3. Signal correction

The calibration factor C_C was corrected for changes in thermochemical states between calibration and measurement as shown in Eq. (13). For the calibration flame, our N_2 -thermometry Raman measurements showed insignificant deviations from adiabatic flame temperatures (within 1% which is smaller than the expected uncertainty from the N_2 -fit). Thus, the adiabatic conditions (1986 K, Han mechanism) are assumed. Due to the slightly larger deviations from the adiabatic conditions for the $\text{NH}_3/\text{H}_2/\text{N}_2$ flames, the temperatures for these all are taken from the N_2 -thermometry Raman measurements. For all conditions, main species are taken from 1D-simulations (Han mechanism). Relative changes in the number density, Boltzmann fraction, dimensionless line overlap, and fluorescence yield are illustrated in Fig. 10 for all flames with 39% H_2 in the fuel to assess the sensitivity of the signals to these parameters. Note that for transition-dependent parameters in Fig. 10(b), only values for the strongest excitation line $Q_1(15.5)$ are shown. Compared to the calibration flame, the number density increases by $\approx 25\%$ for lean mixtures where the temperature is much lower. The Boltzmann fraction scales directly with temperature, and the changes are roughly 10% from very lean to stoichiometric conditions. Boltzmann fractions decrease with lower temperatures because the total population of $v'' = 1$ decreases. Flame temperatures around stoichiometry are almost the same and therefore f_B/f_{B_C} approaches unity. The changes of line overlap Γ due to changes of the broadening environment are in the order of 2%. Fluorescence yield F shows a non-monotonic behavior due to changes in the quenching environment (recall that $F \propto 1/Q$). The NO quenching rate for all flames is dominated by H_2O ($> 80\%$) for all cases. Quenching by O_2 is decreasing from $\approx 15\%$ contribution at $\phi = 0.6$ to 0% at stoichiometric conditions and explains the initial increase of the fluorescence yield on the lean side. On the rich side, the NO self quenching rate decreases, which causes a further increase in fluorescence yield. At very rich conditions, the simulations predict a small amount of ammonia slip and therefore quenching on ammonia appears for $\phi > 1.3$. Due to the interplay of number density, Boltzmann fraction and quenching, the signal correction as a function of the equivalence ratio is slightly larger than one for lean and rich flames and reaches its minimum in the range of approx. -10% in relation to the methane calibration flame under stoichiometric conditions.

4.3.4. NO mole fractions

Figs. 11(a)–(d) show all experimental and numerical results for NO mole fractions in the post flame region of the $\text{NH}_3/\text{H}_2/\text{N}_2$ flames. The corresponding experimental data are available in the supplementary material. The NO mole fractions increase significantly with increasing

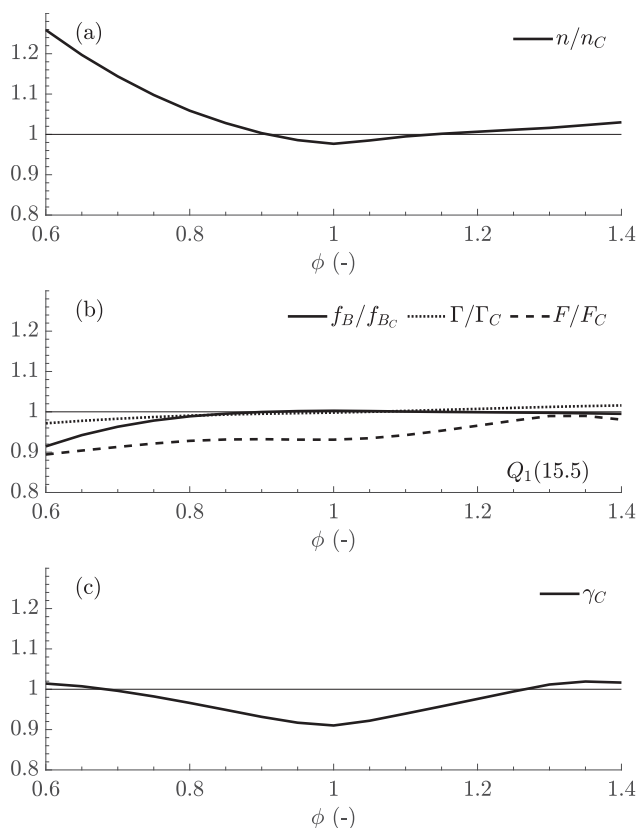


Fig. 10. Relative changes, related to the calibration flame, of number density (a) as well as Boltzmann fraction, dimensionless line overlap, and fluorescence yield (b) for all flames with a fuel content of 39% H_2 as a function of the equivalence ratio. Line-dependent values in (b) are shown for the $Q_1(15.5)$ transition. (c) shows the total correction value as a function of the equivalence ratio that is applied to the signals (see Eq. (13)).

H_2 content in the fuel mixtures. For example, if the fuel contains 45% H_2 , the NO levels are still in the order of 2500 ppm at the richest measured condition of $\phi = 1.4$, while at 27% H_2 there is no NO present. As elaborated by others, NO in ammonia combustion is mainly produced through the HNO-channel [55,57,58]. In the early stage of combustion NH_3 is consumed primarily by reaction with OH and secondarily with H and forms NH_i (NH , NH_2). For almost all fuel mixtures and equivalence ratios, NH_i is then oxidized through an HNO intermediate to NO (HNO-channel). Only in very rich mixtures, the extended Zel'dovich mechanism [59] plays a significant role in NO formation through the abundance of H atoms that are required for the formation of molecular nitrogen from radical attack of H to NH . Especially NO-production routes via the HNO-channel highly depend on the H/O/OH-radical concentrations [14]. These are promoted with increasing the H_2 content in the fuel due to enhanced reactivity. Based on numerical simulations, Fig. 12 shows on the one hand the positive correlation of the radical concentrations of OH, O and H with the NO concentration in the product gases, and on the other hand the increased radical concentrations for higher H_2 ratios in the fuel.

The agreement between experiments and simulations is in general better for fuel rich and far-lean mixtures across all H_2 levels presented in this study. The mechanisms by Nakamura and Stagni show very good agreement with the experiments for rich mixtures and the mechanisms by Wang and Han tend to underpredict NO emissions slightly, especially with increased H_2 content in the fuel. Independent of the chemical mechanism, for lean and stoichiometric mixtures the comparison to experimental data reveal a slight but systematic underprediction of NO mole fractions. This observation is consistent with

the measurements from Hayakawa et al. [15], where the measured NO mole fractions of a binary mixture of 30% H_2 and 70% NH_3 (where $X_{H_2}/X_{NH_3} = 0.428$, which is almost the same ratio as in the 27 % H_2 fuel mixture) are systematically higher than the predicted mole fractions.

Table 4 lists all peak NO concentrations as well as the relative deviations of the numerical simulations from the experiments. NO mole fractions peak around $\phi \approx 0.85 - 0.9$ in all experimental cases, which in general agrees very well with the predictions of all tested mechanisms. However, peak NO mole fraction show deviations of up to 25%, depending on the mechanism. Deviations of the Jiang mechanism are in the order of 40%–50%. The predictive capabilities are naturally related to the number of reactions and species included in the kinetic model (see Table 2) and the conditions to which the model has been optimized. As the Jiang mechanism was optimized for preheated and pressurized conditions, it is expected that its predictive capabilities will be reduced for the present conditions. However, as the Jiang mechanism contains only 64 reactions (Han:130; Wang: 444) it has significant advantages in computational efficiency and the trend of increasing NO is well captured.

The highest measured NO mole fraction is around 1.3% for 45% H_2 at $\phi = 0.85$. Peak NO mole fractions increase with increasing H_2 by a factor of ≈ 1.6 from 27% to 45% H_2 . A highly non-linear increase, as measured for laminar flame velocities [60], is not observed in the present dataset. Significant amounts of NO are present in the 45% H_2 case even at very rich conditions (2360 ppm at $\phi = 1.4$). For lower levels of H_2 , NO emissions decrease significantly earlier. Reduced reactivity for lower levels of H_2 has been shown to be manageable [8,9] and thus our results confirm to keep the decomposition ratio as low as possible to mitigate NO emissions. Further quantitative NO measurements in a larger number of different flames are needed to better understand the primary NO formation in NH_3 -based flames. This is a prerequisite for knowledge-based design of future technical combustion systems.

5. Conclusion

NO mole fractions in the post flame region of laminar $NH_3/H_2/N_2$ -air flames were measured non-intrusively using laser induced fluorescence. The study included four different fuel mixtures with a H_2 content varying from 27% to 45% H_2 over a wide range of equivalence ratios. LIF-signals were corrected for absorption, signal trapping, and pulse energy fluctuations. A calibration approach based on the addition of NO to a lean CH_4 flame was presented. Signal corrections were performed to transfer the calibration to thermochemical conditions differing from the calibration flame. Corrections to account for changes in number density, Boltzmann fraction, line overlap, and electronic quenching were made based on main species concentrations taken from 1-D simulations and experimentally measured temperatures using N_2 -Raman thermometry. Temperature measurements show slight deviations from adiabatic flame temperatures when the flow velocity is tuned to match the laminar flame speed predicted by the Han reaction mechanism. Different sources of experimental uncertainties were analyzed and included in the results, indicating that the largest uncertainty in the present study comes from NO A-state quenching cross sections. Signal corrections based on changes in number density, Boltzmann fraction, absorption line broadening, and quenching were in the order of 10%.

Peak NO mole fractions increase nearly in direct proportion to the amount of H_2 in the fuel mixture. The strong increase of NO mole fractions can be correlated to the increase of the O/H-radical pool. The largest measured NO mole fraction was 1.3% (45% H_2 , $\phi = 0.85$). The tested kinetic models show good agreement for rich and far-lean mixtures. However, all tested kinetic models tend to underpredict NO mole fractions of lean mixtures, with a slight tendency for better agreement at lower levels of H_2 in the fuel mixture. The presented data can be used for the validation and improvement of chemical kinetic models. However, spatially resolved data on NO and intermediate species is required to experimentally characterize local NO formation in the reaction zone.

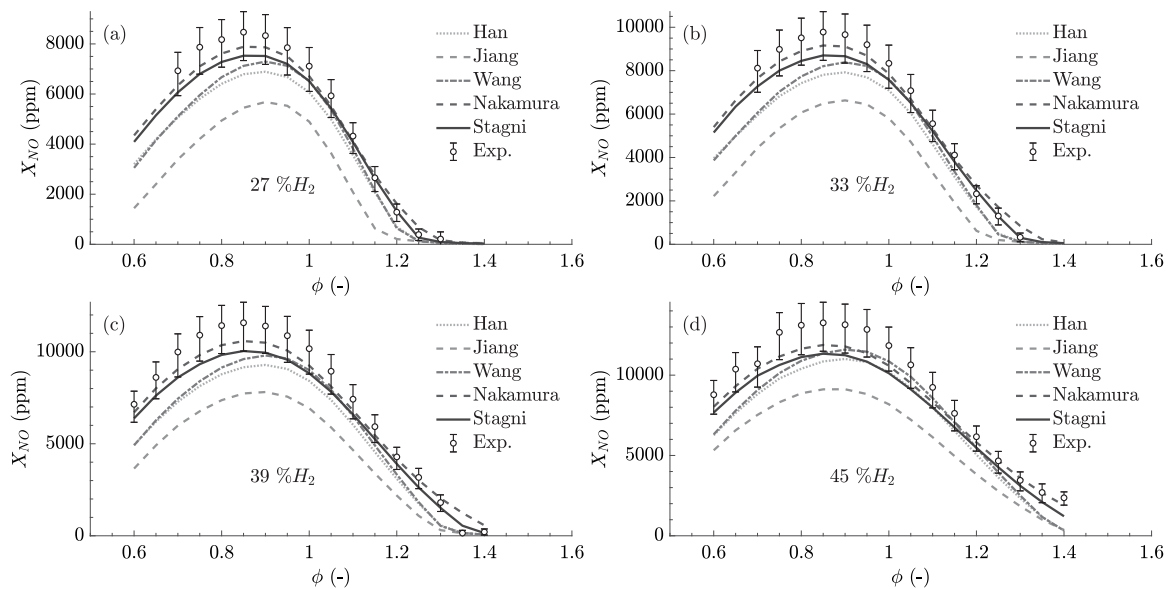


Fig. 11. Experimental and numerical mole fractions of NO for four different $\text{NH}_3/\text{H}_2/\text{N}_2$ blends ranging from 27% to 45% H_2 . Error bars combine stochastic uncertainties (1σ), uncertainties from the calibration measurement (95% confidence interval) and systematic errors as described in 3.4.

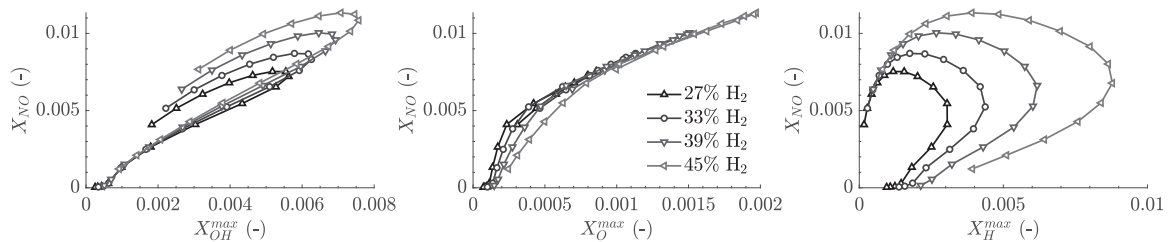


Fig. 12. Correlation of maximum OH, O and H radical concentrations to product NO mole fractions (Stagni mechanism).

Table 4
Peak NO concentrations (ppm).

Blend	Experiment	Han	Wang	Nakamura	Stagni	Jiang
27% H_2	8469	6889 (−23%)	7294 (−16%)	7887 (−7%)	7528 (−12%)	5680 (−49%)
33% H_2	9779	7922 (−23%)	8388 (−17%)	9161 (−7%)	8703 (−12%)	6631 (−47%)
39% H_2	11 575	9283 (−25%)	9797 (−18%)	10580 (−9%)	10040 (−15%)	7812 (−48%)
45% H_2	13 257	11008 (−20%)	11590 (−14%)	11877 (−12%)	11334 (−17%)	9121 (−45%)

CRediT authorship contribution statement

M. Richter: Writing – original draft, Visualization, Methodology, Investigation, Data curation. **J. Lill:** Investigation, Data curation. **R.S. Barlow:** Writing – review & editing, Supervision, Methodology. **A. Gruber:** Writing – review & editing, Supervision. **A. Dreizler:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **J.R. Dawson:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **D. Geyer:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary materials associated with this article can be found in the online version <https://doi.org/10.1016/j.combustflame.2025.114139>

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